

RIMSHA NOREEN

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Objective

Having completed an MS in Mathematics with expertise in graph theory, network analysis, and computational modeling, I am currently pursuing an M2 in Mathematics and Computer Science as a Bézout Scholar. My research focuses on modeling and analyzing complex networks using Python, graph-based methods, and data-science techniques. I aim to pursue a PhD at the intersection of applied mathematics, computer science, and data science, with a focus on real-world networked systems.

Education

Université Gustave Eiffel, Champs-sur-Marne (France) (2025 - ongoing)
M2 Mathematics & Computer Science (Bézout Scholar)
Major Subjects : Algorithms, Programming, Complexity, Probability, Graph theory, Geometry, Optimization, Data science, Advanced Geometry, Advanced Data Science

COMSATS University Islamabad, Lahore Campus (2023 - 2024)
MS Mathematics | CGPA: **4.0 / 4.0**
Major Subjects: Advanced Topics in Graph Theory, Fuzzy Logic and Applications, Mathematical Biology

COMSATS University Islamabad, Lahore Campus (2019 - 2023)
BS Mathematics | CGPA: **3.5 / 4.0** | **4th Rank in Batch**
Major Subjects: Introduction to Computing, Graph Theory, Introduction to Computer Programming, Discrete Mathematics, Linear Algebra, Algorithms, Numerical Computations, Mathematical Statistics, Numerical Optimization, Operations Research

Research Experience & Publications

M2 Research Internship: Exploiting GNNs (Graph Neural Networks) for detecting cyber-attacks in the smart Grid (Ongoing)

- Developing Graph Neural Network (GNN)-based models for detecting False Data Injection (FDI) attacks in smart grid systems.
- Leveraging graph topology and temporal data to identify stealthy cyber-physical attacks
- Designing robust and adaptive machine learning frameworks for real-time anomaly detection
- Working with IEEE benchmark datasets and simulation environments for model validation.
- Investigating model generalization, scalability, and interpretability in graph-based intrusion detection systems.
- Supervisors: **Prof. E. V. Belmega** (UGE-ESIEE Paris) and **Prof. S. Lakshminarayana** (University of Warwick, UK)

MS Thesis: Analyzing Topological Indices for Phenylacetone Monooxygenase Network Using Curve Fitting Model

- Conducted research in graph theory focused on the computation of topological indices for the PAMO (Phenylacetone Monooxygenase) network.
- Developed both manual and Python-based methods for efficient computation and data analysis of topological indices.
- Applied curve fitting models to identify structural patterns and relationships within the network data.
- Integrated computational techniques with mathematical modeling for network analysis.
- Demonstrated proficiency in graph theory, network analysis, and Python programming.

BS Project: Zero Forcing Number for Biswapped Networks

- Investigated the zero forcing number for biswapped networks (BSN), including path, cycle, and complete graph structures.
- Applied advanced graph theory techniques to analyze network properties and forcing dynamics.
- Utilized concepts such as color change rules and forcing sequences for modeling complex graph structures.
- Demonstrated strong skills in mathematical modeling, graph theory, and network analysis.

Published paper: Exploring Topological Indices and Entropy Measures of Phenylacetone Monooxygenase by Using Python Coding

- **Authors:** Rashad Ismail, Rimsha Noreen, Muhammad Farhan Hanif, Muhammad Faisal Hanif, Muhammad Kamran Siddiqui
- **Journal:** Journal of Molecular Graphics and Modelling, 137, 108981, 2025
- **Funding:** Supported by King Khalid University, Grant No. RGP2/245/45.
- **Abstract:** In this study, I computed topological indices and edge weight-based entropy of the Phenylacetone Monooxygenase (PAMO) network using Python programming. The research involved developing and applying computational techniques to analyze the structural and informational properties of the PAMO network. This work contributes to understanding the molecular structure of PAMO and its implications for biocatalysis and drug development.
- DOI: <https://doi.org/10.1016/j.jmglm.2025.108981>

Submitted paper : Exploring Topological Indices and Entropy Measures for Iron Nitride Network via Python and Logarithmic Regression

Abstract: In this research, I analyzed the structural properties of the Iron Nitride network by computing topological indices and entropy measures using Python. I utilized logarithmic regression to model and interpret complex data patterns, showcasing my expertise in computational techniques and advanced mathematical modeling. This work emphasizes my proficiency in data-driven analysis for understanding complex networks.

Technical Skills

Programming Languages: C, C++, Python, MATLAB

Research & Computational Tools: Mathematica, VS Code, LaTeX, MATLAB, IBM SPSS statistics, Maple, Desmos graphing calculator, Sagemath, Tora

Data Science & Analytical Skills: Graph Theory, Network Analysis, Data Analysis, Curve Fitting, Statistical Modeling, Machine Learning Fundamentals

Research Areas: Graph Neural Networks (GNNs), Complex Networks, Computational Modeling, Smart Grid Security, Applied Graph Theory

Professional Skills: Scientific writing, Research Design, Technical Report Writing, Academic Presentation, Microsoft Office (Word, PowerPoint, Excel)

Certifications and Achievements

Python and Artificial Intelligence Course Completion

Successfully completed a course hosted by the Microsoft Learn Student Ambassadors program focusing on Python and AI concepts.

Webinar Participation: "Introduction to Graph Database"

Participated in a webinar exploring graph theory's role in structuring and analyzing complex data relationships, with a focus on data science and machine learning applications.

Certificate of Appreciation

(December 21, 2022)

Awarded by the Department of Mathematics, COMSATS University Islamabad, Lahore Campus, for effective and committed efforts in organizing the "Welcome & Farewell Party."

"MATLAB" – Certificate of Completion, Great Learning Academy

(July 2022)

Successfully completed an online MATLAB course covering basic programming, matrix operations, data analysis and visualization using MATLAB.

Certificate of Participation

(December 5, 2023)

Recognized by COMSATS University Islamabad, Lahore Campus, for active participation in the "One Day International Symposium on Mathematical Sciences".

Certificate of Participation

(May 31, 2023)

Awarded by the University of Central Punjab for attending the seminar on "Smart Writing: Leveraging Mathpix and ChatGPT for Effective LaTeX Code."

Online Course Completion: "Simple Coding" – OpenLearn, The Open University

(June 6, 2025)

Successfully completed an online introductory programming course on OpenLearn, developing foundational skills in coding logic, algorithms, and problem-solving.
